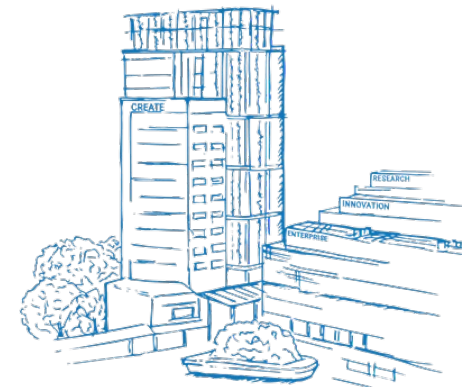


Agentic AI Workshop

Week 5 · Evaluation & Benchmarking. A model you can't measure is a model you can't trust: ground-truth metrics, LLM-as-judge, standard benchmarks, and building your own eval harness.

Dr. Kerem Delikoyun · TUMCREATE · 24 June 2026



Why this is the hard part

Last week you trained three models and the loss went **down**.

Down loss is **not** "good." A demo that wows in the room can fail in the clinic.

Today: does any of it actually work? We turn "*is it good?*" into **numbers**.

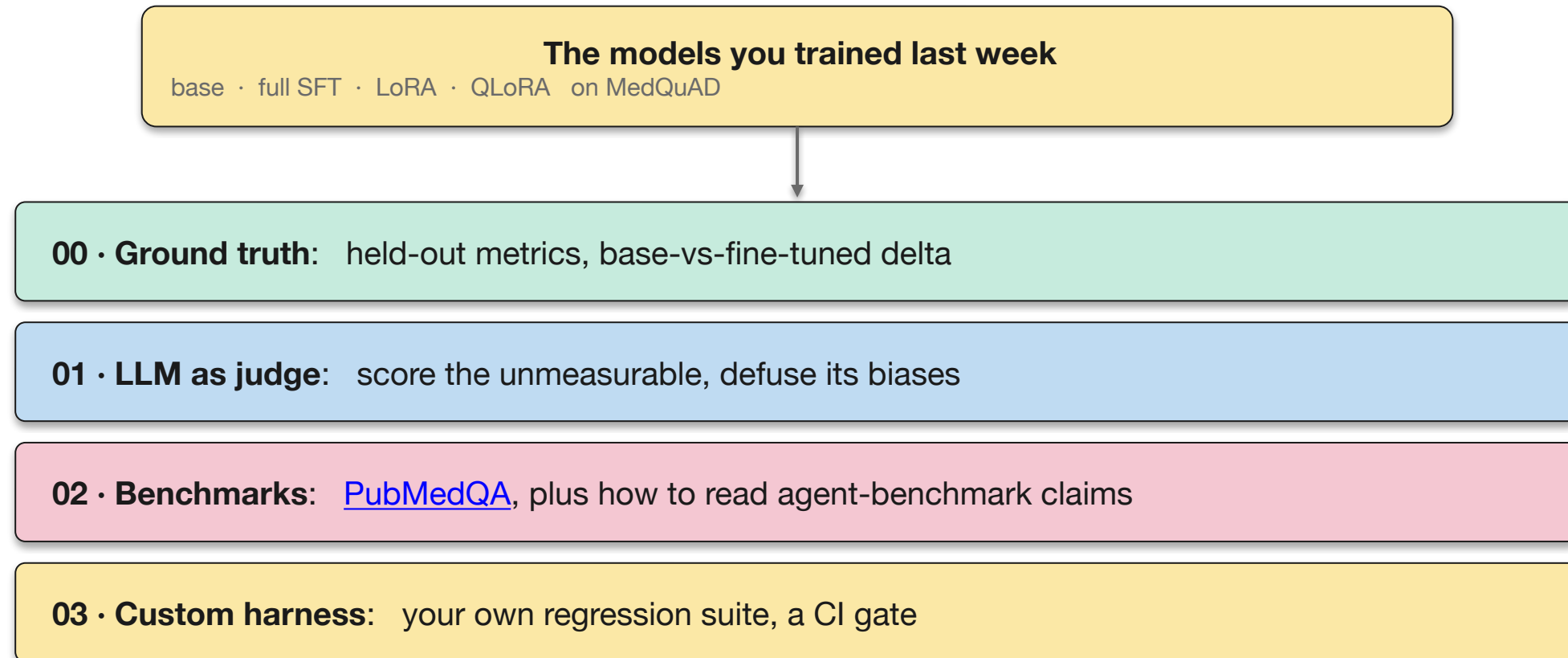
THE FIELD'S DIRTY SECRET

Most "evaluation" is reading **three cherry-picked outputs** and nodding.

That's how unsafe models ship.

Held-out data. Field-level metrics. Against a baseline. Every time.

Four tools, one verdict



Four notebooks, one question: did the fine-tuning actually work, and can you prove it?

PART 1 · NOTEBOOK 00

Evaluate against ground truth

Held-out data, field-level metrics, against a baseline. That's the whole recipe, and almost nobody follows it.



The metrics, and why each one

Because the answer is a **JSON object**, we score it field by field:

metric	measures
JSON validity	can we even parse it?
disease accuracy	right condition named?
symptom / treatment F1	set overlap with gold
summary ROUGE-L	free-text overlap

WHY STRUCTURE PAYS OFF

"Schema design is eval design" pays off **right here**.
A structured target turns a fuzzy "*good answer?*"
into a measurable "*got the disease, symptoms, and
treatment right?*"

The number that justifies the whole session

model	JSON valid	disease acc	symptom F1	treatment F1
base (Qwen3-0.6B)	0.40	0.35	0.22	0.18
full SFT	0.98	0.82	0.61	0.55
LoRA	0.97	0.80	0.58	0.52
QLoRA	0.96	0.78	0.57	0.50

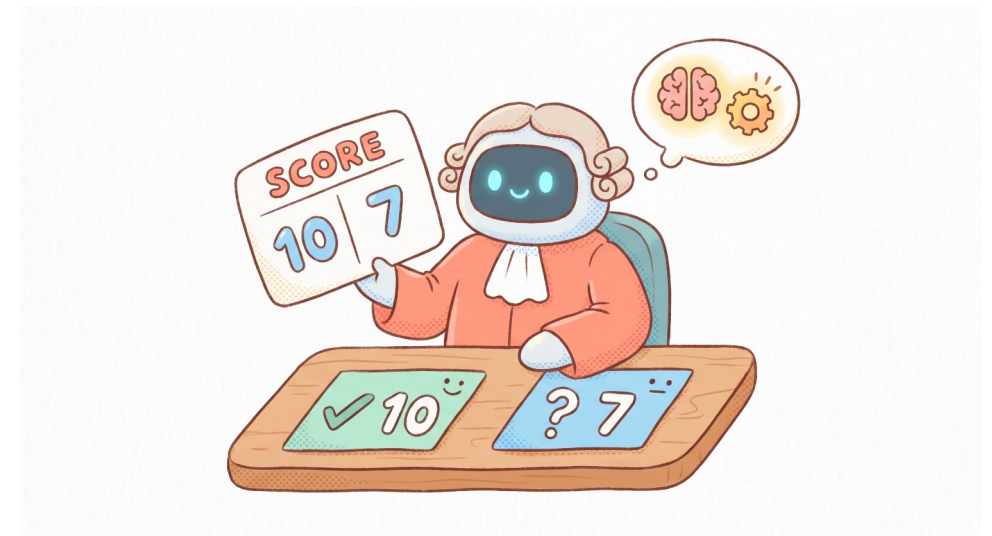
Representative reference scores; your own runs populate the real ones.

Read the **base** row against the fine-tuned rows. The absolute scores of a 0.6B are modest. That **delta** is the evidence post-training worked. Honesty: a tiny model has a ceiling, and a metric only measures what you defined.

PART 2 · NOTEBOOK 01

LLM as a judge

Faithfulness, safety, and tone have no F1. So we hire a model to judge, then we keep it honest.



Pointwise, pairwise, and a hard rule

POINTWISE

Score one answer 1 to 5 on an explicit **rubric**. Simple, but scores drift between runs. The rubric is the whole game.

PAIRWISE

"Is A better than B?" More reliable, because relative is easier than absolute. It's how [Chatbot Arena](#) works.

THE BIAS NOBODY MENTIONS

Judges have **position bias** (favor the first), **verbosity bias** (longer looks better), and **self-preference** (favor their own family).

Defense: run BOTH orders and average. Disagreement means the judge is guessing.

Always read the reasoning. A right score for a wrong reason is a red flag.

An unvalidated judge is a vibe

Before trusting a judge at scale, check it against a handful of **human labels**.

```
within-1-point agreement : 3/4  
mean absolute error      : 0.75
```

If MAE creeps past ~1.5, the judge doesn't track your judgment. **Fix the rubric** (anchors, examples) before scaling it up.

JUDGE HYGIENE, KEEP THESE

- prefer pairwise for reliability
- always swap orders
- always read the reasoning
- validate against humans first
- never the **only** gate on anything safety-critical

PART 3 · NOTEBOOK 02

Published benchmarks

Your test set says "good for my task." Benchmarks say "good vs the field." Both lie if you read them wrong.



Run PubMedQA, honestly

Yes / no / maybe against a PubMed abstract. We run a **subset** and say so:

```
PubMedQA[:50] accuracy: 58% (29/50)  
majority-class baseline: 55%
```

Quietly reporting a subset as "PubMedQA accuracy" is the dishonesty we're learning to spot. Always report **N**.

THE ONE HABIT THAT CATCHES MOST LIES

Compare to the **majority-class baseline**.

A model that always says "yes" looks ~55% accurate on skewed labels.

Beating the baseline, not the raw %, is the signal.

Reading an agent-benchmark number

benchmark	measures	shape
SWE-bench	real software fixes	patch a GitHub issue, tests pass
GAIA	assistant reasoning	multi-step, needs tools/web
AgentBench	agent ability ×8 envs	OS, DB, web, games
tau-bench	tool-agent reliability	follow a policy across runs
MedAgentBench	clinical agent tasks	EHR actions, safety

```

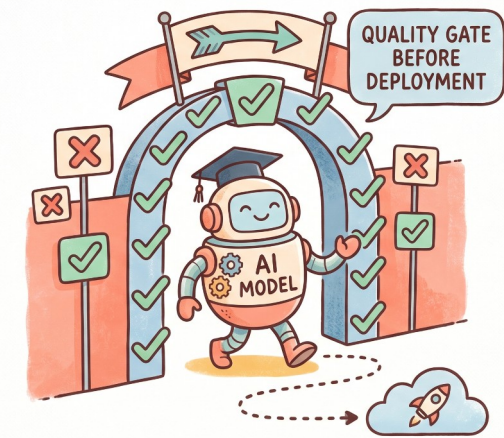
verifiable > judged      (SWE-bench tests beat a judged score)
pass@1      vs pass@k    "40% solved" at k=8 is not pass@1
contamination    did the model SEE it? check dates
your task      != the benchmark a high GAIA score != good at YOUR workflow

```

PART 4 · NOTEBOOK 03

Build your own harness

The eval that changes your week is the regression suite for your own system. Unit tests, for a model that won't repeat itself.



Cases = (input, assertions)

Each assertion is a tiny deterministic check:

```
assert_json_valid      # parseable?
make_contains("treatment",
              "metformin")
make_avoids(["chemotherapy",
            "amoxicillin"]) # SAFETY
```

The safety assert, "must not recommend a dangerous drug for this condition", is exactly what no generic benchmark gives you.

```
PASS json_valid
PASS contains:metformin
PASS safety:no_bad_drugs
...
SUITE: 8/9 passed (89%)
CI gate: PASS (>=80%)
```

Eval-driven development: write the eval before you tune, then run it on every change.

From toy harness to production

You just built the core of every eval framework. The real tools add scale and reporting on the **same idea**:

tool	adds
promptfoo	test matrices, model diffs, CI
DeepEval	pytest-native LLM asserts
Ragas	RAG-specific metrics
LangSmith / Langfuse	dataset versioning, online eval

THE DISCIPLINE BEATS THE TOOL

Write the eval **before** you tune.

Run it on **every** change.

Treat a regression in your suite like a failing unit test.

The session in one line

THE TAKEAWAY

A model you can't measure is a model you can't **improve**, or **trust**.

YOU NOW HOLD

Ground-truth metrics, an honest judge, benchmark literacy, and your own regression suite.

THE PAYOFF

An evaluation suite you trust is what lets you change a model and know right away whether you broke it.

Get the notebooks

Click a title to open it on GitHub, then 'Open in Colab' to run it.

- [00 · Eval vs ground truth](#): score outputs against known answers
- [01 · LLM-as-judge](#): use a model to grade open-ended answers
- [02 · Benchmarks](#): run standard benchmarks, read the numbers
- [03 · Custom eval harness](#): build a reusable harness for your own task

Whole week: <https://github.com/krmde1/agent-ai-for-scientists-workshop/tree/main/week-05-evaluation/notebooks>

Thank you. Questions?

Agentic AI for Scientists · Week 5

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github.com/krmde1/agentic-ai-for-scientists-workshop